

A computationally efficient deep learning model for real-time image dehazing on edge devices

Gavin Halford, Arthur C. Depoian II, and Colleen P. Bailey

Department of Electrical Engineering, University of North Texas, Denton, TX, USA 76207

Abstract—Reliable image restoration in particulate environments is essential for high-stakes applications such as robotic surgery and autonomous navigation. However, existing dehazing models often struggle with a “computational bottleneck,” requiring massive parameter counts that are unsuitable for edge deployment. This paper introduces CED-RIDE, an ultra-lightweight architecture containing only 24,659 parameters. By utilizing channel-separated skip connections and the Lion optimizer, the model addresses the non-uniform scattering and spectral distortion characteristic of real-world smoke.

Evaluated on the SMOKE dataset, CED-RIDE achieves a validation PSNR of 20.11 dB and an SSIM of 0.8405, outperforming existing state-of-the-art benchmarks by 1.28 dB in PSNR and 35% in structural fidelity. These results demonstrate that high-quality, real-time smoke removal can be achieved with a minimal computational footprint, bridging the gap between theoretical restoration and operational edge-computing requirements.

Index Terms—Image Enhancement, Image Dehazing, Smoke Removal, Edge AI

I. INTRODUCTION

The reliability of autonomous computer vision systems is fundamentally governed by the clarity of the transmission medium. Adverse environments, ranging from dense industrial smoke to complex surgical fields, introduce particulate scattering that severely compromises spectral and spatial information [1]. In clinical settings, surgical smoke generated during procedures can obscure tissue visibility and degrade image quality, which directly impacts surgical decision-making and limits the effectiveness of downstream automated image analysis [2]. Additionally, in environmental and security sectors, obstructions like dust and fog significantly heighten accident risks in land transportation, necessitating real-time enhancement and IoT-centric alert systems capable of providing reliable visual data to drivers [3].

Many systems attempt to generalize these problems under the umbrella of “dehazing,” though smoke presents a unique set of challenges. Unlike the relatively uniform distribution of atmospheric haze, smoke particles possess distinct properties and localized concentrations that require specialized detection and removal techniques [4].

The physical degradation of images in these environments is typically described by the atmospheric scattering model:

$$I(x) = J(x)t(x) + A(1 - t(x)) \quad (1)$$

where $I(x)$ represents the observed hazy image, $J(x)$ is the latent scene radiance to be recovered, $t(x)$ is the transmission map, and A is the global atmospheric light [5]. Foundational heuristics, such as the Dark Channel Prior (DCP), have long

been used to estimate the transmission map by observing local pixel intensities [5]. However, traditional methods based on this model often suffer from artifacts or lack robustness when faced with non-uniform pollution gradients or robust sparse representations of haze density [6].

To overcome the limitations of heuristic models, the field has transitioned toward deep learning architectures, including Convolutional Neural Networks (CNNs) and Vision Transformers [7]. Recent efforts have explored more complex inputs, such as 4-channel hazy image generation, to better utilize regional haze properties and improve restoration quality [8]. Despite these advancements, a significant performance gap remains between laboratory benchmarks and field deployment. Recent comparative analyses demonstrate that many algorithms optimized for synthetic datasets fail to maintain consistency when applied to real-world hazy images characterized by unpredictable atmospheric gradients [9]. Furthermore, modern high-fidelity approaches, such as those utilizing Neural Radiance Fields (NeRF) for reliable dehazing, often struggle with the computational complexity of handling complex lighting scenes in real-time [10]. Similarly, unsupervised adaptive enhancement models, while reducing the need for manual priors, often require multi-stage processing that is difficult to accelerate for high-resolution video applications on the edge [11].

The necessity to advance the field lies in the development of “parameter-lean” architectures that do not sacrifice structural clarity for speed. While lightweight U-Net models have attempted to replace computationally complex DCP operations with convolutional layers, there is a persistent need for even more efficient refinement strategies for mobile edge deployment [12]. Building upon the principles of efficient AI and feature-level enhancement established in our previous work on low-light recovery [13], we propose a computationally efficient deep learning model for real-time image dehazing on edge devices, **CED-RIDE**.

Our work bridges the gap between high-fidelity restoration and operational feasibility. By implementing channel-separated skip connections, specifically designed to address the wavelength-dependent scattering common in particulate smoke, our model achieves a PSNR of 20.11 dB and an SSIM of 0.8405. With a footprint of only 24,659 parameters, CED-RIDE provides a robust solution for real-time smoke removal on resource-constrained edge hardware, effectively addressing the efficiency bottleneck identified in current literature.

II. METHODOLOGY

The SMOKE Dataset consists of 220 training images, 110 pairs, each containing a clean, original image alongside one featuring a dense or non-uniform particle distribution generated using a fog machine [14]. Additionally there are 12 image pairs for testing or validation. Each image is size 640×512 in size. Example clean and noisy paired images from both the training and testing set are shown in Figure 1.



Fig. 1: SMOKE Dataset [14]

The proposed convolutional architecture introduces channel-separated skip connections in parallel to provide guided filtering to each channel individually. At the same time, an RGB fusion path steps through two layers of expansive filters before being condensed into a reduction layer, preparing it for concatenation with the resultants of each channel filtering path. This information is then exposed to two progressive 2D convolution layers fusing the information into a final cleaned image output. The complete system architecture is summarized in Figure 2.

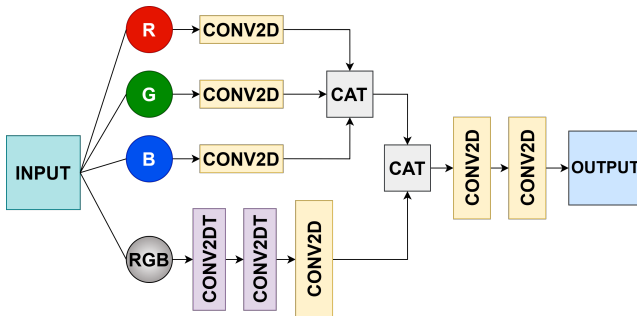


Fig. 2: Smoke Removal System Architecture

The training configuration for the proposed smoke removal and image dehazing model was designed to balance performance and efficiency, particularly for potential deployment in real-time or edge-based applications. The model contains a total of 24,659 trainable parameters, making it lightweight

and well-suited for resource-constrained environments. Optimization was carried out using the Lion optimizer, selected for its adaptive capabilities and convergence speed. The learning rate was set to 0.002, with a small weight decay of 0.00005 to prevent overfitting and promote generalization. Training was conducted over 100 epochs, and the total training time was approximately 2 minutes and 31 seconds, highlighting the model's efficiency during the learning phase.

To guide the training process, a custom loss function was employed that combines both perceptual quality and pixel-level fidelity. Specifically, the loss was formulated as a weighted combination of the Structural Similarity Index Measure (SSIM) and the inverse of the Peak Signal-to-Noise Ratio (PSNR), expressed as

$$\alpha \cdot \text{SSIM} + (1 - \alpha) \cdot \frac{1}{\text{PSNR}}.$$

This formulation ensures that the model prioritizes structural clarity and perceptual quality while still maintaining low reconstruction error. The weighting factor α was empirically set to 0.8, placing greater emphasis on SSIM during training. This configuration was instrumental in achieving high-quality haze-free image restoration while preserving critical scene details.

III. RESULTS

TABLE I: Quantitative comparison on the SMOKE dataset [14].

Method	PSNR↑	SSIM↑
DCP	11.26	0.26
GDN	15.19	0.53
MSBDN	13.19	0.34
DeHamer'22	13.31	0.28
Jin et al.	18.83	0.62
CED-RIDE	20.11	0.8405

The performance of CED-RIDE was evaluated on the non-uniform SMOKE dataset [14], with quantitative results compared against traditional and state-of-the-art (SOTA) methods in Table I. Our model achieved a validation PSNR of 20.11 dB and an SSIM of 0.8405, outperforming the best-performing baseline, Jin et al., by 1.28 dB. Notably, the model provides a significant improvement in structural fidelity over traditional methods like the Dark Channel Prior (DCP) [5], which yielded an SSIM of only 0.26 due to the complex, non-uniform nature of smoke scattering.

The training dynamics, shown in Figure 3, illustrate the rapid and stable convergence facilitated by the Lion optimizer. With only 24,659 parameters, the architecture efficiently maps smoky features to clear scenes without the computational overhead of large-scale transformers. Figure 4 and Figure 5 confirm that the validation metrics reached a steady state without oscillations or overfitting. While the training set reported a PSNR of 18.81 dB and an SSIM of 0.84191, the validation PSNR of 20.11 dB indicates effective generalization to unseen configurations. Maintaining an SSIM above 0.84 across both

phases suggests that the channel-separated skip connections reliably preserve edge and texture information compared to heavier models like MSBDN (0.34 SSIM).



Fig. 3: Training

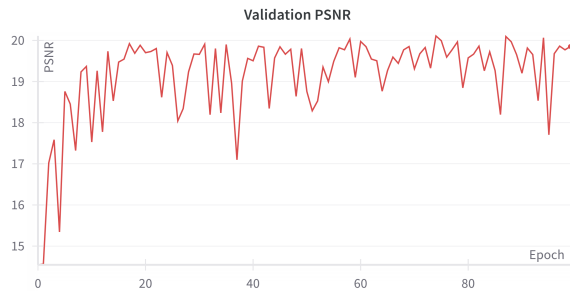


Fig. 4: PSNR Validation Data

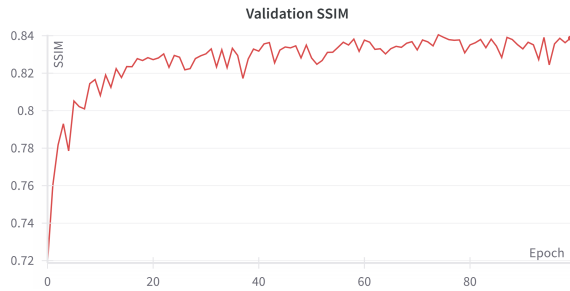


Fig. 5: SSIM Validation Data

IV. CONCLUSION

The proposed deep learning model demonstrates effective haze removal from images while maintaining a highly efficient architecture with only 24,659 parameters. This compact design makes the model particularly well-suited for real-time applications where computational resources are limited. Such efficiency is especially valuable in critical scenarios including firefighter vision enhancement in smoke-filled environments, improved visibility for autonomous vehicles navigating hazy conditions, and strengthened surveillance capabilities for defense and security purposes. The model achieves strong performance metrics, with a validation PSNR of 20.11 dB

and an SSIM of 0.8405, reflecting both high fidelity and perceptual quality. Moreover, the training process, powered by the Lion optimizer, converges in remarkably little time, further supporting its practicality for rapid development and deployment. Future work will focus on adapting and deploying this model on edge devices to fully leverage its real-time potential in operational settings.

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